

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID: SIH26187

Organization: Ministry of Home Affairs

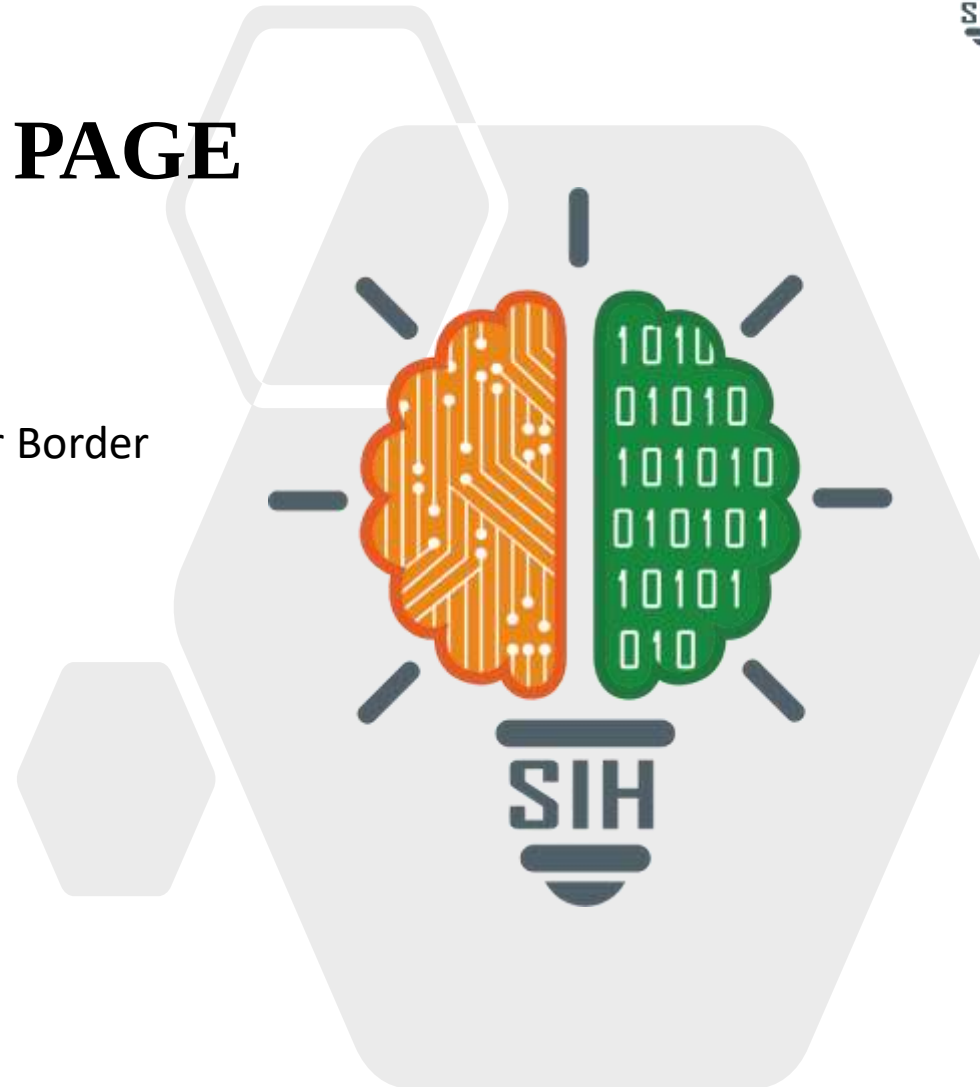
Title: AI-Based Intelligent Video Analytics Platform for Border Surveillance using Existing CCTV Infrastructure

Category: Software

Domain: Blockchain & Cybersecurity

AI-Powered Intelligent Border Surveillance System

Team Name: Sentrix





Proposed Solution (Describe Idea/Solution/Prototype)



AI-Powered Surveillance

Upgrades passive CCTV into an active AI surveillance system that detects, tracks and analyses suspicious activity in real time, cutting the need for manual monitoring.



Multi-Target Detection

Detects people, vehicles, animals, restricted-zone intrusions and vehicle number plates; assigns each target a tracking ID and follows its movement across the feed.



Auto-Flagged Alerts

Every intrusion is auto-flagged with an evidence-backed alert, so a restricted-zone breach reaches the operator dashboard within seconds instead of being buried in raw footage.



Unique & Explainable AI

Uses a unique detection model that unifies tracking, geofencing, number-plate intelligence and provides an Explainable Threat Score into one dashboard.

Real-time Detection Feed ● Live



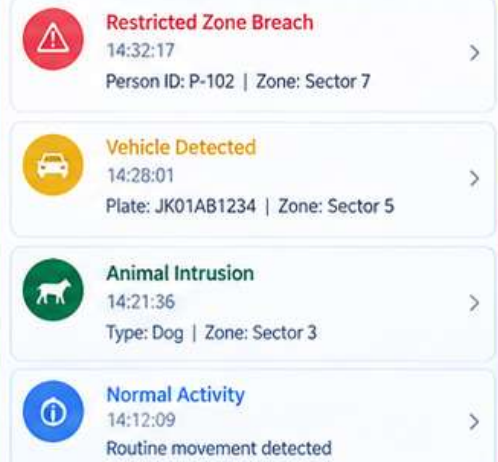
Threat Analytics Dashboard



Border Map View



Alerts & Logs





TECHNICAL APPROACH

DETECT • TRACK • ANALYZE • PROTECT



| 1. Technology Stack

**Frontend & Dashboard**
HTML5
CSS3
JavaScript (ES6)
Responsive Command Centre UI

**AI / ML Engine**
TensorFlow.js
COCO-SSD Object Detection
Browser-based inference
Real-time detection

**Analytics & Visualization**
Chart.js
Threat Score Analytics
Activity Trends
Incident Statistics

**Mapping & Geofencing**
Leaflet.js
Border Zone Mapping
Restricted Areas
Geofence Detection

| 2. AI Processing Pipeline



| 3. Threat Score Engine

**Threat Score =**

- Location
- Time
- Zone Entry
- Repeat Activity
- Behaviour



| 4. Architecture Advantages

**Real-Time Processing**

**Modular Architecture**

**Edge-Friendly Deployment**

**Scalable System**

 | **From Camera Feed to Actionable Intelligence — In Seconds.**



1. Technical Feasibility

Built with proven and accessible technologies

- ✓ **AI/ML Models** – Uses YOLO for detection, reliable and lightweight for real-time inference.
- ✓ **Web Tech** – HTML5, CSS3, JavaScript (ES6) for smooth and responsive UI.
- ✓ **Hardware** – Works with standard cameras and edge devices (Raspberry Pi / Jetson Nano for future scale).
- ✓ **Scalable** – Modular architecture allows easy integration of more zones, sensors and AI models.



2. Economic Feasibility

Cost-effective and resource efficient

-  **Low Development Cost**
Uses open-source tools and existing hardware.
-  **Minimal Hardware Requirement**
Runs on standard cameras and local servers.
-  **Low Maintenance**
Simple UI and modular system reduces upkeep.
-  **High ROI**
Prevents security breaches, reduces manpower costs and saves resources in the long run.



3. Social Feasibility

Acceptable, useful and easy to adapt

-  **Supports Security Forces**
Helps border personnel with real-time alerts and actionable insights.
-  **Builds Public Trust**
Increases safety and reduces illegal activities.
-  **Easy to Use**
Simple interface and clear alerts.
-  **Scalable for Wider Use**
Can be extended to other sensitive areas (wildlife, airports, urban zones).



4. Market & Commercial Viability

High demand, strong growth potential



- 40%** Government & Defence
Border and internal security
- 25%** Law Enforcement
Police, paramilitary, intelligence
- 20%** Industrial & Private Sector
Critical infrastructure, smart cities
- 15%** Global Market
Export potential for AI surveillance solutions



5. Long-Term Impact

Safer borders, stronger nations

-  **Reduced Border Infiltration**
-  **Faster Response Time**
-  **Better Resource Utilization**
-  **Safer & More Secure Communities**

**Feasible Today.
Valuable Tomorrow.**

BORDER SIGHT AI is not just a project —
it's a step towards a safer, smarter
and more secure future.

IMPACT AND BENEFITS

DETECT • TRACK • ANALYZE • PROTECT



Impact on Target Audience

Border Security & Surveillance Operators



Before

- Hours of manual CCTV monitoring
- Operator fatigue
- Delayed intrusion detection



With BORDER SIGHT AI

- Real-time AI detection
- Evidence-backed alerts
- Faster response time



Social Impact

Safer Borders & Checkpoints



Faster identification of suspicious activity



Reduced undetected intrusions



Improved security for nearby communities



Better situational awareness for security forces



Economic Benefits

Cost-Effective Surveillance

Traditional
Monitoring

High manpower cost



BORDER SIGHT AI



Automated monitoring



Key Benefits

- ✓ Uses existing CCTV cameras
- ✓ Browser-based AI reduces infrastructure costs
- ✓ Reduces dependency on 24/7 manual monitoring
- ✓ Modular system reduces maintenance cost



Environmental Benefits

Smarter Wildlife Monitoring

Animal
DetectedAI
ClassificationAvoid False
Security AlertsReduced
Unnecessary
Patrols

Key Benefits

- Distinguishes wildlife from potential threats
- Reduces unnecessary patrol deployment
- Minimizes disturbance in sensitive border ecosystems

**BORDER SIGHT AI transforms passive surveillance into faster, smarter and actionable border intelligence.**

Faster Response



Improved Security



Lower Cost



Sustainable Monitoring



1. Prototype & Source Code

BORDER SIGHT AI Prototype

- ✓ Interactive AI surveillance dashboard
- ✓ Built using modern web technologies
- ✓ Modular and scalable architecture



[Yashhh710/BorderSight-AI](https://github.com/Yashhh710/BorderSight-AI)



Live Demo

[BorderSight AI – Secure Login](#)



2. AI Detection Research

AI / ML Technology



TensorFlow.js

- Browser-based machine learning



COCO-SSD

- Real-time object detection
- Detects people, vehicles and animals



3. Visualization & Mapping

Analytics & Geospatial Intelligence



Chart.js

- Threat analytics
- Activity trends
- Incident visualization



Leaflet.js

- Border mapping
- Geofencing
- Restricted-zone visualization



4. Domain Research & Validation

Border Surveillance Research

- ✓ Studied perimeter-surveillance approaches
- ✓ Analyzed geofencing concepts
- ✓ Designed restricted-zone intrusion logic
- ✓ Developed explainable threat-scoring methodology



Technology Ecosystem



HTML5 + CSS3 + JavaScript



TensorFlow.js



COCO-SSD



Chart.js



Leaflet.js



Built on proven technologies and validated through modern AI surveillance concepts.